HW 4: Theory

Friday, October 11, 2024 4:04 PM

1. Assume X is a discrete random variable that takes values in $\{1, 2, 3\}$, with probability defined by

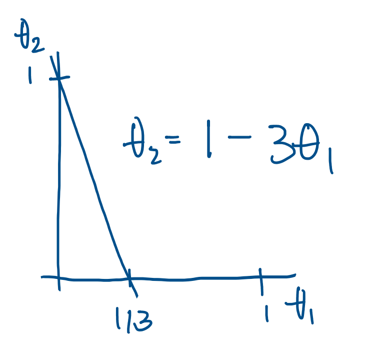
$$\Pr(X = 1) = \theta_1$$
$$\Pr(X = 2) = 2\theta_1$$
$$\Pr(X = 3) = \theta_2,$$

where $\theta = [\theta_1, \theta_2]$ is an unknown parameter to be estimated.

Now assume we observe a sequence $D := \{x^{(1)}, x^{(2)}, \ldots, x^{(n)}\}$ that is *independent and identically distributed* (*i.i.d.*) from the distribution. We assume the number of observations of the values: $1, 2, 3$ in D are s_1, s_2, s_3, respectively.

(a) **[5 points]** To ensure that $\Pr(X = i)$ is a valid probability mass function, what constraint should we put on $\theta = [\theta_1, \theta_2]$? Write your answers quantitatively as expressions that include θ_1 and θ_2.

$$3\theta_1 + \theta_2 = 1$$
$$0 \leq \theta_1$$
$$0 \leq \theta_2$$



(b) **[5 points]** Write down the joint probability of the data sequence

$$\Pr(D \mid \theta) = \Pr\left(\left\{x^{(1)}, \ldots, x^{(n)}\right\} \mid \theta\right),$$

and the log probability $\log \Pr(D \mid \theta)$.

$$L(\theta) = P[\{x^{(1)}, x^{(2)}, \ldots, x^{(n)}\} \mid \theta] = \theta_1^{s_1} \cdot (2\theta_1)^{s_2} \cdot \theta_2^{s_3}$$
$$= \theta_1^{s_1} \cdot 2^{s_2} \cdot \theta_1^{s_2} \cdot \theta_2^{s_3}$$
$$= 2^{s_2} \theta_1^{s_1 + s_2} \theta_2^{s_3}$$
$$= 2^{s_2} \theta_1^{s_1 + s_2} (1 - 3\theta_1)^{s_3}$$

$$\ell(\theta) = \ln\left(2^{s_2} \theta_1^{s_1 + s_2} (1 - 3\theta_1)^{s_3}\right)$$
$$= s_2 \ln(2) + (s_1 + s_2) \ln \theta_1 + s_3 \ln(1 - 3\theta_1)$$

(c) **[5 points]** Calculate the maximum likelihood estimation $\hat{\theta}$ of θ based on the sequence D.

$$\frac{\partial}{\partial \theta_1} \ell(\theta) = \cancel{0} + \frac{s_1 + s_2}{\theta_1} + \frac{s_3}{1 - 3\theta_1}(-3)$$

$$0 = (s_1 + s_2)(1 - 3\theta_1) - 3s_3\theta_1$$

$$0 = s_1 + s_2 - 3s_1\theta_1 - 3s_2\theta_1 - 3s_3\theta_1$$

$$0 = s_1 + s_2 - 3\theta_1(s_1 + s_2 + s_3)$$
$$0 = \frac{1}{3}\left(\frac{s_1 + s_2}{s_1 + s_2 + s_3}\right) - \theta_1$$

$$\nabla'_{\theta_1} \ell(\theta) = \nabla_{\theta_1}\left((s_1 + s_2)\theta_1^{-1} + s_3(1 - 3\theta_1)^{-1}\right)$$
$$= -\frac{1}{2}(s_1 + s_2)\theta_1^{-2} +$$

$$\boxed{\hat{\theta}_1 = \frac{1}{3} \cdot \frac{s_1 + s_2}{s_1 + s_2 + s_3} \qquad \hat{\theta}_2 = 1 - \frac{s_1 + s_2}{s_1 + s_2 + s_3} = \frac{s_3}{s_1 + s_2 + s_3}}$$

second derivative test:

$$\frac{\partial^2}{\partial \theta_1^2} \ell(\theta) = -\frac{s_1 + s_2}{\theta_1^2} - \frac{s_3}{(1 - 3\theta_1)^2}(-3)(-3)$$

$$= -\frac{s_1 + s_2}{\theta_1^2} - \frac{9s_3}{(1 - 3\theta_1)^2} < 0$$

$\Rightarrow L(\theta)$ concave, i.e. we're found a max

2. **[10 points]** Let $\{x^{(1)}, \ldots, x^{(n)}\}$ be an *i.i.d.* sample from an exponential distribution, whose the density function is defined as

$$f(x \mid \beta) = \frac{1}{\beta} \exp\left(-\frac{x}{\beta}\right), \quad \text{for} \quad 0 \leq x < \infty.$$

Please find the maximum likelihood estimator (MLE) of the parameter β. Show your work.

$$f(x \mid \beta) = \frac{1}{\beta} \exp\left(-\frac{x}{\beta}\right), \quad 0 \leq x < \infty$$

$$L(\beta) = \prod_{i=1}^n \frac{1}{\beta} \exp\left(-\frac{x_i}{\beta}\right)$$
$$= \beta^{-n} \prod_{i=1}^n \exp\left(-\frac{x_i}{\beta}\right)$$
$$= \beta^{-n} \exp\left(-\frac{\sum x_i}{\beta}\right)$$
$$\ell(\beta) = -n \ln(\beta) - \beta^{-1} \sum_{i=1}^n x_i$$
$$\frac{\partial}{\partial \beta} \ell(\beta) = -n\beta^{-1} + \beta^{-2} \sum_{i=1}^n x_i$$
$$0 \overset{!}{=} \beta - \frac{1}{n} \sum_{i=1}^n x_i$$
$$\boxed{\hat{\beta} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}}$$

$$\frac{\partial^2}{\partial \beta^2} \ell(\beta) = n\beta^{-2} - 2\beta^{-3} \sum x_i$$
$$= n\beta^{-3}(\beta - 2\bar{x})$$
$$< 0 \text{ when } \beta < 2\bar{x}$$
$$\beta = \hat{\beta} = \bar{x} < 2\bar{x} \Rightarrow \hat{\beta} \text{ is a max.}$$

3. (a) **[10 points]** Assume that you want to investigate the proportion (θ) of defective items manufactured at a production line. You take a random sample of 30 items and found 5 of them were defective. Assume the prior of θ is a uniform distribution on $[0, 1]$. Please compute the posterior of θ. It is sufficient to write down the posterior density function upto a normalization constant that does not depend on θ.

$$\theta \sim \text{uniform } (0, 1), \quad p(\theta) = 1, \quad 0 \leq \theta \leq 1$$

$$y \sim \text{bin}(30, \theta), \quad p(y \mid \theta) = \binom{30}{k} \theta^k (1 - \theta)^{30 - k}$$

$$p(\theta \mid y = 5) \propto p(y = 5 \mid \theta) \cdot p(\theta)$$
$$= \binom{30}{5} \theta^5 (1 - \theta)^{25} \cdot 1$$
$$\boxed{\propto \theta^{6-1} (1 - \theta)^{26-1} \Rightarrow \theta \sim \text{beta}(6, 26)}$$

(b) **[10 points]** Assume an observation $D := \{x^{(1)}, \ldots, x^{(n)}\}$ is *i.i.d.* drawn from a Gaussian distribution $\mathcal{N}(\mu, 1)$, with an unknown mean μ and a variance of 1. Assume the prior distribution of μ is $\mathcal{N}(0, 1)$. Please derive the posterior distribution $p(\mu \mid D)$ of μ given data D.

$$\mu \sim N(0, 1), \quad p(\mu) \propto \exp\left(-\frac{1}{2}\mu^2\right)$$
$$x \sim N(\mu, 1), \quad p(x \mid \mu) \propto \exp\left(-\frac{(x - \mu)^2}{2}\right)$$
$$p(D \mid \mu) \propto \prod_{i=1}^n \exp\left(-\frac{(x_i - \mu)^2}{2}\right)$$
$$= \exp\left(-\frac{\sum (x_i - \mu)^2}{2}\right)$$

$$x \sim N(\mu, \sigma^2$$
$$f(x) = (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right)$$

$$p(\mu \mid D) \propto p(D \mid \mu) \cdot p(\mu)$$
$$\propto \exp\left(-\frac{1}{2}\sum (x_i - \mu)^2\right) \cdot \exp\left(-\frac{1}{2}\mu^2\right)$$
$$= \exp\left(-\frac{1}{2}\left(\sum (x_i - \mu)^2 + \mu^2\right)\right)$$
$$= \exp\left(-\frac{1}{2}\left(\sum (x_i^2 - 2\mu x_i + \mu^2) + \mu^2\right)\right)$$
$$= \exp\left(-\frac{1}{2}\left(\sum x_i^2 - 2\mu \sum x_i + (n+1)\mu^2\right)\right)$$
$$\propto \exp\left(-\frac{1}{2}\left((n+1)\mu^2 - 2\mu \sum x_i\right)\right)$$
$$\propto \exp\left(-\frac{n+1}{2}\left(\mu^2 - 2\mu \frac{\sum x_i}{n+1}\right)\right)$$
$$\propto \exp\left(\frac{1}{2\left(\frac{1}{n+1}\right)}\left(\mu - \frac{\sum x_i}{n+1}\right)^2\right)$$

$$\Rightarrow \boxed{\begin{aligned} &\mu \mid D \sim N\left(\mu = \frac{\sum x_i}{n+1}, \sigma^2 = \frac{1}{n+1}\right) \\ &p(\mu \mid D) = \left(2\pi\left(\frac{1}{n+1}\right)\right)^{-1/2} \cdot \exp\left(-\frac{1}{2}\left(\frac{1}{n+1}\right)^{-1}\left(\mu - \frac{\sum x_i}{n+1}\right)^2\right) \\ &\qquad = (2\pi)^{-1/2}(n+1)^{1/2} \exp\left(-\frac{1}{2}(n+1)\left(\mu - \frac{\sum x_i}{n+1}\right)^2\right) \end{aligned}}$$

By Baye's Rule we can simply identify the kernel of a prob. distribution and assume the normalizing constants will work themselves out, i.e. the pdf will integrate to 1. This is how I am able to add and remove constant factors while doing algebra above.